



List of 40 mm grenades

This is a general collection of the world's many types of ammunition for grenade launchers in 40 mm (1.57 in) caliber.

Several countries have developed or adopted grenade launchers in 40 mm caliber.^[1]

NATO

NATO currently uses three standardized 40 mm grenade families: 40 mm low velocity (LV), 40 mm medium velocity (MV), and 40 mm high velocity (HV). Low- and medium-velocity cartridges are used for different hand-held grenade launchers, while the high-velocity cartridge is used for automatic grenade launchers.

40×46 mm LV (40 mm low velocity)

40×46 mm LV (low velocity)^[2] is a NATO-standard^[3] high-low grenade launcher cartridge meant for hand-held grenade launchers, such as the M79, M203, Milkor MGL, Heckler & Koch AG36 and M320 Grenade Launcher Module.

The propellant has low pressure and gives the projectile an average velocity of 78–84 m/s (256–276 ft/s) depending on the ammunition type.^[4]

40 mm low-velocity ammunition types (NATO)



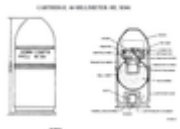
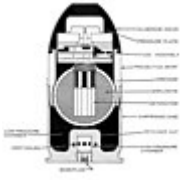
Box full of 40×53 mm high-velocity grenades



A 40×46 mm low-velocity training round being fired from an M203 grenade launcher



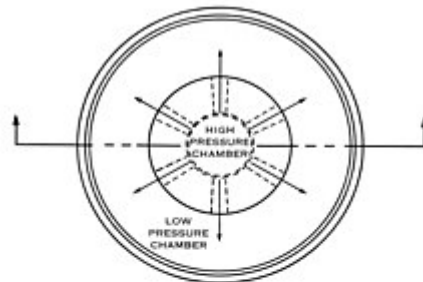
A 40×46 mm low-velocity cartridge being loaded into an M203 grenade launcher attached to an M16A1

HE, high explosive	
M381 <i>HE, high explosive</i> ^[5] Basic high-explosive shell. ^[5]	
M386 <i>HE, high explosive</i> ^[6] Basic high-explosive shell. ^[6]	Missing image
M406 <i>HE, high explosive</i> ^[7] Basic high-explosive shell. ^[7]	
M441 <i>HE, high explosive</i> ^[8] Basic high-explosive shell. ^[8]	Missing image

AB, air burst	
M397, M397A1 <i>AB, air burst</i> ^{[9][10]} Octol filled fragmentation grenade with a time fuze. ^[9] The A1 has a different fuze from the regular M397. ^[10]	Missing image

DP, dual purpose	
M433 <i>HEDP, high-explosive dual-purpose</i> ^[11] Shaped charge with ability to damage soft targets and penetrate armor. ^[11] Armor penetration: 2.5 inches (63 millimeters). ^[12]	

High-low system casing for the 40×46 mm low-velocity cartridge



WHEN IGNITED BY THE PRIMER, THE PROPELLANT ENCLOSED IN THE BRASS POWDER CHARGE CUP (HIGH PRESSURE CHAMBER) RUPTURES THE CUP AT THE VENT HOLES AND IS VENTED INTO THE REMAINDER OF THE CARTRIDGE CASE (LOW PRESSURE CHAMBER).

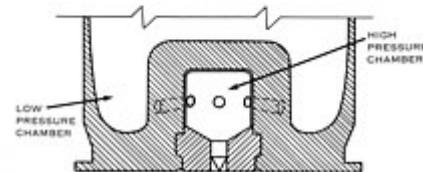


Figure 21. High-low propellant system.



Fired 40 mm low-velocity M781 showing its orange signal chalk

MP, multiple projectile	
M576 MP-APERS, multiple projectile anti- personnel Buckshot cartridge with twenty 24-grain bullets. ^[13]	Missing image

IL, illumination	
M583A1 Illumination/signal flare, parachute star ^[14] (white flare) Flare grenade with parachute for illumination and signaling. ^[14]	 M583A1
M585 Illumination/signal flare, cluster star ^[15] (white flare) Grenade containing five nose-ejected, free-falling pyrotechnic star pellets used for signaling. ^[15]	Missing image
M661 Illumination/signal flare, parachute star ^[14] (green flare) Flare grenade with parachute for illumination and signaling. ^[14]	Missing image
M662 Illumination/signal flare, parachute star ^[14] (red flare) Flare grenade with parachute for illumination and signaling. ^[14]	Missing image
M992 Infra-red illumination flare Flare grenade with parachute for infrared illumination.	 COMM TEL M992

S, smoke	
M676 <i>Smoke canopy (yellow smoke)</i> ^[16] Smoke grenade with parachute for signaling and marking. ^[16]	Missing image
M680 <i>Smoke canopy (white smoke)</i> ^[17] Smoke grenade with parachute for signaling and marking. ^[17]	Missing image
M682 <i>Smoke canopy (red smoke)</i> ^[18] Smoke grenade with parachute for signaling and marking. ^[18]	Missing image
M713 <i>Ground marker (red smoke)</i> Smoke grenade for signaling and marking. ^[19]	Missing image
M714 <i>Ground marker (white smoke)</i> Smoke grenade for signaling and marking.	Missing image
M715 <i>Ground marker (green smoke)</i> ^[20] Smoke grenade for signaling and marking. ^[20]	Missing image
M716 <i>Ground marker (yellow smoke)</i> ^[21] Smoke grenade for signaling and marking. ^[21]	Missing image

G, gas	
M651 <u>CS gas</u> ^[22] <u>Tear gas</u> <u>grenade</u> . ^[22]	Missing image

TB, thermobaric	
XM1060 <u>Thermobaric</u> ^[23] The XM1060 is a 40 mm thermobaric grenade developed by <u>Picatinny Arsenal</u> . ^[23]	Missing image

P, practice	
M781 <u>P, practice</u> ^[24] Training ammunition consisting of a steel body filled with orange signal chalk capped with a plastic <u>ogive cap</u> . ^[24]	

Besides combat ammo there also exists crowd control ammunition like sponge grenades.

40 mm low-velocity ammunition types (Sweden)

Sweden currently operates the M203 grenade launcher (designated *Granattillsats 40 mm Automatkarbin* in Sweden) and thus uses the 40 mm low-velocity cartridge.^{[25][26][4]} Going against Swedish military tradition, the 40 mm low-velocity cartridge currently lacks a specified indigenous designation in Swedish service.^{[26][4]} Instead only the projectile types have designations.

Currently these projectile types can be found in Swedish service manuals.^[4]



HE, high explosive

40 GSGR

HE, high explosive

Name: *40 GSGR*, abbreviation for *40 mm gevärsspränggranat*, Swedish for 40 mm rifle high-explosive grenade. The fuze is designated *ÖHKBAR 40 GSGR*, abbreviation for *ögonblickligt högkänsligt basanslagsrör* + shell designation, Swedish for 'instant high sensitivity base impact fuze'.^{[26][4]}

Description: The type is a high-explosive grenade meant against soft targets.

Construction: The shell consists of a fragmentation body outfitted with internal steel balls for extra fragmentation damage. A highly sensitive, zero-delay fuze is located at the bottom of the shell.^{[26][4]}

Marking: The shell has a yellow head and green body. The side of the shell features the marking "ST" in yellow, which stands for *stålkulor* ('steel balls').^{[26][4]}

40 GSGR data	
Cartridge weight	0.264 kg (0.58 lb)
Shell weight	0.19 kg (0.42 lb)
Primary charge	3 g (0.11 oz) <u>PETN</u>
Main charge	23 g (0.81 oz) <u>RDX</u>
Propellant	0.35 g (0.012 oz) <u>NCGL</u>
High pressure	70 MPa
Low pressure	15 MPa
Muzzle velocity	84 m/s (280 ft/s)
<u>NEM</u> ^[a]	26 g (0.92 oz)
Sources	^{[26][4]}

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DP, dual purpose

40 GPSGR

HEDP, high-explosive dual-purpose

Name: *40 GPSGR*, abbreviation for *40 mm gevärspansarspränggranat*, Swedish for "40 mm rifle high explosive anti-tank grenade".^{[26][4]}

Description: The type is a high-explosive dual-purpose grenade meant for use against both soft and lightly protected targets. It is probably equivalent to the American 40 mm M433.^[11]

Construction: The shell consists of a fragmentation body, a high-explosive charge and a shaped charge.^{[26][4]}

Marking: The shell has a green head and green body. The side of the shell features the marking "RSV" in yellow, which stands for *riktad sprängverkan* ("shaped charge", lit. "directed explosive effect").^{[26][4]}

40 GPSGR data

Cartridge weight	0.26 kg (0.57 lb)
Shell weight	0.19 kg (0.42 lb)
Primary charge	0.33 g (0.012 oz) CH-6
Main charge	43.5 g (1.5 oz) Comp A5 ^[b]
Propellant	0.34 g (0.012 oz) <u>NCGL</u>
High pressure	70 <u>MPa</u>
Low pressure	15 <u>MPa</u>
Muzzle velocity	78 m/s (260 ft/s)
<u>NEM</u> ^[c]	44.3 g (1.6 oz)
Sources	^{[26][4]}

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P, Practice

40 GÖVNGR 07

P, practice

Name: *40 GÖVNGR 07*, abbreviation for *40 mm gevärsövningsgranat 07*, Swedish for 40 mm rifle practice grenade 07.^{[26][4]}

Description: The type is a practice shell for combat training and practice shooting.

Construction: The shell has a steel body with a plastic cap and is filled with red signal chalk.^{[26][4]}

Marking: The shell is colored in NATO blue training color.^{[26][4]}

40 GÖVNGR 07 data

Cartridge weight	0.26 kg (0.57 lb)
Shell weight	0.19 kg (0.42 lb)
Primary charge	-
Main charge	Red signal chalk
Propellant	0.47 g (0.017 oz) NCGL
High pressure	70 MPa
Low pressure	15 MPa
Muzzle velocity	78 m/s (260 ft/s)
NEM^[d]	5 g (0.18 oz)
Sources	^{[26][4]}

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Mockups and inert types also exist for loading exercises and educational purposes.

40 mm low-velocity ammunition types (Romania)

Romanian arms producer ROMARM has made a version of their 40 mm rifle-mounted grenade launcher AG-40 chambered in 40×46 mm NATO (then designated AG-40PN). Production of Romanian 40 mm low-velocity ammunition is handled by the arms factory *Uzina Mecanica Ploeni*,^[27] a subsidiary of ROMARM. The projectiles seem to be of Romanian origin based on available information.^[27]

 HE, high explosive

Grenade 40 NATO Exploziva

HE, high explosive

Description: Romanian 40×46 mm low-velocity high-explosive cartridge.^[27]

Construction: The shell is made of steel and has a point fuze. The main explosive charge is located at the bottom of the shell. Below the main explosive charge is a layer of metal balls for extra fragmentation damage.^[27]

Grenade 40 NATO Exploziva data	
Cartridge length	112 mm (4.4 in)
Cartridge weight	370 g (13 oz)
Shell weight	245 g (8.6 oz)
Muzzle velocity	75 m/s (250 ft/s)
Lethal radius	10 m (33 ft)
Self-destruct	16-23 s
Sources	^[27]

Missing image

 P, practice

Grenade 40 NATO Inerta

P, practice

Name: The cartridge is named *Inerta* (inert), but the cartridge is live. *Inert* refers to the projectile, which is a solid material projectile.^[27]

Description: Romanian 40×46 mm low-velocity practice cartridge.^[27]

Construction: The shell is solid in construction and made of duralumin.^[27]

Grenade 40 NATO Exploziva data	
Cartridge length	112 mm (4.4 in)
Cartridge weight	370 g (13 oz)
Shell weight	245 g (8.6 oz)
Muzzle velocity	75 m/s (250 ft/s)
Lethal radius	-
Self-destruct	-
Sources	^[27]

Missing image

SAGM fuze

The United States Army Armament Research, Development and Engineering Center (ARDEC) began development of a 40 mm smart airburst fuze (proximity fuze) in 2011 to improve the ability of grenade launchers like the M203 and M320 to engage targets in defilade. Called *small arms grenade munitions* (SAGMs), they double the lethality of the standard M433 grenade round by adding a small "smart" fuze sensor that detonates in the air to hit targets in cover or behind obstacles. The airburst function is similar to the XM25 CDTE, which has an onboard laser system to determine the distance to the target, but SAGMs are considered complementary to the XM25 rather than competing against it, as the XM25 provides low-angle fire while 40 mm launchers fire a lobbing trajectory. Integrated sensors and logic devices scan and filter the environment and then autonomously airburst the fuze without needing to be told to by the firer, thereby not requiring the soldier to carry extra weapon accessories. SAGMs enable soldiers to accurately incapacitate personnel targets in defilade at ranges between 50 and 500 meters. The round is engineered with three firing modes: airburst; point detonation; and self-destruct. A successful demonstration occurred in November 2013.^[28] Although the SAGM sensor does not need a laser rangefinder or any pre-fire programming sequence, it does require some skill by the user to aim and fire the round correctly so that it can detect the wall or obstruction to detonate in the air. The SAGM was to undergo evaluation in July 2015 and, if successful, transition into an official Army Program of Record by the end of the year.^[29] Not only does the fuze burst over walls, but it can detonate when passing cover like trees, bursting just as it senses and passes the trunk. The sort of sensor SAGMs use to differentiate clutter from triggering obstacles is highly classified, but shows airburst reliability of 76 percent.^[30]

40×51 mm MV (40 mm medium velocity)

40×51 mm MV (medium velocity),^[2] also known as *40×51 mm extended range low pressure (ERLP)*,^[31] is a NATO-standard^{[32][33]} high–low grenade launcher cartridge meant for hand-held grenade launchers. Its purpose is to be an intermediate cartridge between the 40×46 mm low-velocity and 40×53 mm high-velocity cartridges and is thus referred to as 40 mm medium velocity.^[2]

The propellant has medium pressure and gives the projectile an average velocity of 100 m/s (328 ft/s) depending on the ammunition type.^[2] It has a maximum range of 800 meters, exceeding conventional extended range low-velocity variants by up to 375 meters.^[2]

The 40×51 mm MV cartridge was designed by Rheinmetall Denel Munitions for the US Special Operations Command (USSOCOM) after a 2008 requirement for enhanced range and lethality from hand-held 40 mm grenades.^[34] Rheinmetall answered by developing a new family of 40 mm grenades named 40 mm medium velocity^[34] and by 2019 the cartridge was undergoing NATO qualification.^[32]

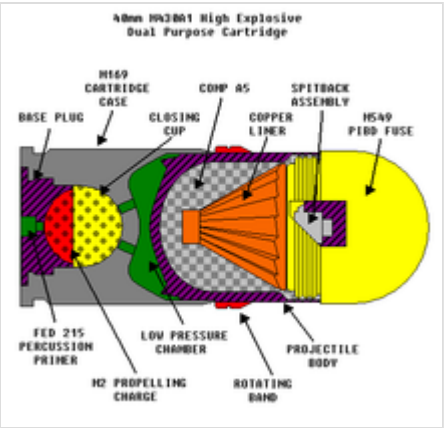
Besides NATO the cartridge has been ordered by the South African National Defence Force (SANDF) as the cartridge for their next generation multiple grenade launcher, the Milkor Y4.^{[2][31]} SANDF approved acquisition in February 2018 but deliveries could not be finished until the end of 2020 due to the COVID-19 pandemic.^[31]

40×53 mm HV (40 mm high velocity)

40×53 mm HV (*high velocity*)^[2] is a NATO-standard^[3] high–low grenade launcher cartridge meant for mounted or crew-served automatic grenade launchers, such as the Mk.19 AGL, Mk 47 Striker, HK GMG, STK 40 AGL, and Daewoo K4.

The propellant has high pressure and gives the projectile an average velocity of 241 m/s (791 ft/s) depending on the ammunition type.^[4]

40 mm high-velocity ammunition types (NATO)



40 mm high-velocity high-explosive dual-purpose M430A1 cartridge

HE, high explosive	
M383 <i>HE, high explosive</i>^[35] High explosive shell filled with composition A5.^[35] (Comp 5A = hexogen + graphite according to Swedish manuals).^[4] Wound radius: 15 meters. Arms: 18-36 meters^[36]	Missing image
M384 <i>HE, high explosive</i>^[37] High explosive shell filled with composition A5.^[35] (Comp 5A = hexogen + graphite according to Swedish manuals).^[4]	Missing image



40×53 mm high-velocity ammunition belt

DP, dual purpose	
M430, M430A1 <i>HEDP, high-explosive dual-purpose</i>^[38] M430: Shaped charge with ability to damage soft targets and penetrate armor. It arms within 18 to 30 meters of the gun muzzle and has a point-initiating, base-detonating (PIBD) fuze. Armor penetration: 2 inches (51 millimetres).^[38] of steel armor at 0-degree obliquity Inflict personnel casualties: 15 meters from impact^[39] M430A1: Has a longer shaped charge than the M430 and penetrates more armor. Armor penetration: 3 inches (76 millimetres).^[38]	

CA, canister	
M1001 <i>HVCC, high-velocity canister cartridge</i>^[40] Canister shot containing several flechettes. Produces a 3 to 4 ft (0.91 to 1.2 m) wide dispersion pattern at 50 m (164 ft).^[40]	Missing image

AB, air burst	
XM1176 <i>HEDP-AB,</i> <i>high-explosive</i> <i>dual-purpose</i> <i>air burst</i>^[41] Dual purpose shaped charge with programmable fuze for air burst functionality.^[41]	 The image shows a cylindrical XM1176 HEDP-AB air burst round. It has a green body with a black band around the middle and a gold-colored nose. The label 'XM1176' is visible at the bottom of the image.
MK285 <i>PPHE/SD,</i> <i>programmable</i> <i>prefragmented</i> <i>high</i> <i>explosive/self-</i> <i>destructible</i> The <u>MK285</u> is an anti- personnel cartridge designed for the <u>Mk 47</u> <u>Striker</u> automatic grenade launcher. It consists of an electronic programmable fuze, a pre- fragmented warhead and a propulsion system. The fuze is programmed through the fire control of the gun. The fuze is mechanically armed at approximately 23 meters. The round is programmed to airburst over the target and the fuze counts down the programmed time via its built in electronics. If an unprogrammed round is fired, it will detonate on impact. The projectile has a built in self-	Missing image

destruct and can be fired by any automatic grenade launcher.	
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P, practice	
M385, M385A1 <i>P, practice</i> ^{[42][43]} M385: Training round featuring a solid metal projectile. ^[42] M385A1: Updated M385 featuring an ogive equal to the M430 HEDP round. ^[43]	Missing image
M918 <i>P, practice</i> . ^[44] The M918 is a training practice cartridge that has the same muzzle velocity of 790 feet per second (fps), signature, and sound as the HE round. ^[44] ^[45]	Missing image
M922, M922A1 Dummy round ^{[46][47]} Trainers use dummy rounds to check weapon function and to train crews. ^[48]	Missing image
MK281 Mod 0, MK281 Mod 1 <i>P, practice</i> ^[49] MK281 Mod 0: Training round featuring an impact marker. ^[49] MK281 Mod 1: Training round featuring a day-night marker. ^[49]	Missing image

40 mm high-velocity ammunition types (Sweden)

Sweden currently operates the Mk 19 grenade launcher (designated *40 mm granatspruta 92* in Sweden) and thus uses the 40 mm high-velocity cartridge.^{[50][26][4]} Going against Swedish military tradition, the 40 mm high-velocity cartridge currently lacks a specified indigenous designation in Swedish service.^{[26][4]}

Instead only the projectile types have designations.

Currently these projectile types can be found in Swedish service manuals.^[4]

 **DP, dual purpose**

40 PSGR
HEDP, high-explosive dual-purpose
Name: *40 PSGR*, abbreviation for *40 mm pansarspränggranat*, Swedish for 40 mm high explosive anti-tank grenade. The fuze is designated *ÖHKSAR PSGR*, abbreviation for *ögonblickligt högkänsligt spetsanslagsrör* + shell designation, Swedish for instant high sensitivity point impact fuze.^{[26][4]}
Description: The type is a high-explosive dual-purpose grenade meant for use against both soft and lightly protected targets.^{[26][4]} It is probably equivalent to the American 40 mm M430.^[38]
Construction: The shell consists of a fragmentation body, a high-explosive charge, and a shaped charge. The fuze is a highly sensitive fuze without delay that is located at the top of the shell.^{[26][4]}
Marking: The shell has a gold head and green body. The side of the shell features the marking "RSV" in yellow, which stands for *riktad sprängverkan* ("shaped charge", lit. "directed explosive effect").^{[26][4]}

40 PSGR data	
Cartridge weight	0.34 kg (0.75 lb)
Shell weight	0.25 kg (0.55 lb)
Primary charge	0.3 g (0.011 oz) Comp A3 ^[e]
Main charge	33 g (1.2 oz) Comp A5 ^[f]
Propellant	4.45 g (0.16 oz) <u>NCGL</u> M2
High pressure	286 MPa
Low pressure	94 MPa
Muzzle velocity	241 m/s (790 ft/s)
Shaped charge penetration	50 mm (2 in) of steel at 0 degrees obliquity at any range. ^{[38][51]}
<u>NEM</u> ^[g]	38 g (1.3 oz)
Sources	^{[26][4]}



40 PSGR data	
Cartridge weight	0.34 kg (0.75 lb)
Shell weight	0.25 kg (0.55 lb)
Primary charge	0.3 g (0.011 oz) Comp A3 ^[e]
Main charge	33 g (1.2 oz) Comp A5 ^[f]
Propellant	4.45 g (0.16 oz) <u>NCGL</u> M2
High pressure	286 MPa
Low pressure	94 MPa
Muzzle velocity	241 m/s (790 ft/s)
Shaped charge penetration	50 mm (2 in) of steel at 0 degrees obliquity at any range. ^{[38][51]}
<u>NEM</u> ^[g]	38 g (1.3 oz)
Sources	^{[26][4]}





P, practice

40 ÖVNGR

P, practice

Name: 40 ÖVNGR, abbreviation for 40 mm övningsgranat, Swedish for "40 mm practice grenade".^{[26][4]}

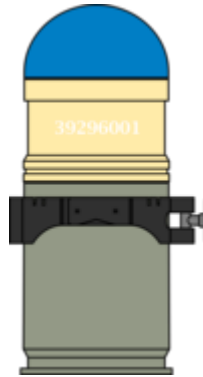
Description: The type is a practice shell for combat training and practice shooting.^{[26][4]} It is probably equivalent to the American 40 mm M918.^[44]

Construction: The shell has a flash charge instead of a combat charge.^{[26][4]}

Marking: The shell is colored in NATO blue training color at the top.^{[26][4]}

40 ÖVNGR data

Cartridge weight	0.37 kg (0.82 lb)
Shell weight	0.25 kg (0.55 lb)
Primary charge	—
Main charge	1.8 g (0.063 oz) flash charge
Propellant	3.5 g (0.12 oz) NC A/S 0200
High pressure	286 MPa
Low pressure	94 MPa
Muzzle velocity	241 m/s (790 ft/s)
NEM^[h]	5.6 g (0.20 oz)
Sources	^{[26][4]}



40 ÖVNGR 07

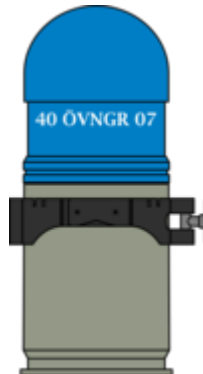
P, practice

Name: 40 ÖVNGR 07, abbreviation for 40 mm övningsgranat 07, Swedish for "40 mm practice grenade 07".^{[26][4]}

Description: The type is a practice shell for combat training and practice shooting.^{[26][4]}

Construction: The shell has a steel body with a plastic cap and is filled with orange signal chalk.^{[26][4]}

Marking: The shell is colored in NATO blue training color. The side of the shell features a marking of the cartridges designation in white.^{[26][4]}



40 ÖVNGR 07 data	
Cartridge weight	0.37 kg (0.82 lb)
Shell weight	0.25 kg (0.55 lb)
Primary charge	-
Main charge	Orange signal chalk
Propellant	4 g (0.14 oz) <u>NCGL</u>
High pressure	286 MPa
Low pressure	94 MPa
Muzzle velocity	241 m/s (790 ft/s)
<u>NEM</u> ^[i]	4.8 g (0.17 oz)
Sources	[26][4]

Mockups and inert types also exist for loading exercises and educational purposes.

Green ammunition

The MK281 is a new type of 40 mm target practice grenade ammunition that has been accepted for use into the United States Marine Corps and the United States Army. It is "green" because it is non-toxic and non-dud producing (since it is a training round), meaning that there is no unexploded ordnance left to clean up on the range and heavy metals in the fuze do not leak into the ground. The MK281 was introduced into parts of the U.S. Armed Forces because of an executive order mandating that they buy green ammunition. The MK281 is manufactured by an American subsidiary of the Rheinmetall Group.

The United States Army has a requirement for a non-dud producing 40 mm training ammunition in both high- and low-velocity variants. The Army awarded four contracts to three United States companies to test designs. The resulting ammunition will not contain explosive energetics and have day and night visible, infrared, and thermal signatures.

Other

40×47 mm (Poland)

40×47 mm is a cartridge caliber produced in Poland for their Pallad wz. 74 rifle-mounted grenade launchers (used with the AK family of rifles in the Polish Army, like the AKM/AKMS, Tantal and Beryl) and Pallad-D wz. 83 grenade launcher (standalone variant fitted with standard pistol grip and folding stock from the AKMS assault rifle). The construction is similar to the one used in 40×46 mm grenades, but they are not interchangeable.



Pallad-D wz. 83

40×47 mm (Romania)

40×47 mm is a cartridge caliber produced in Romania for their AG-40 model 77 and model 80 (today AG-40P) rifle-mounted grenade launchers.^[27] It features a casing with a high–low system. The propellant has low pressure and gives the projectile an average velocity of 78–120 m/s (256–394 ft/s) depending on the ammunition type.^[27]



PA md. 86 assault rifle with 40 × 47 mm AG-40 grenade launcher

Production was originally handled by the arms factory *Uzina Mecanica Filiasi*, however production was later moved to the arms factory *Uzina Mecanica Tohan Zărnești*,^[27] today more commonly known as *S. Tohan S.A.*,^[52] a subsidiary of ROMARM.

Several types of the Romanian 40×47 mm exist:

- High explosive^[27]
- High-explosive fragmentation^[27]
- Smoke^[27]
- Incendiary^[27]
- Tear gas^[27]
- Practice, featuring a small flash charge and smoke signal.^[27]
- Inert^[27]

Tohan currently (2021) offers a 40×47 mm high explosive type called GETZ (*Grenadă Explozivă Tohan Zărnești*) and an inert version called GITZ (*Grenadă Inertă Tohan Zărnești*). Both cartridges are 105 mm (4.13 in) long, with GETZ weighing 0.260 kg (0.573 lb) and GITZ 0.200 kg (0.441 lb).^{[27][52]}

40×74.5 mm (Romania)

40×74.5 mm is a cartridge caliber produced in Romania for their AGA-40 Model 85 automatic grenade launcher.^[27] It features a casing with a high–low system. The propellant has high pressure and gives the projectile an average velocity of 216–223 m/s (709–732 ft/s) depending on the ammunition type.^[27]

Production is handled by the arms factory *Uzina Mecanica Plopeni*,^[27] a subsidiary of ROMARM.

Three ammunition types are known:



An AGA-40 grenade launcher

- A high-explosive grenade producing 150 fragments weighing 0.2 g (0.0071 oz) each, creating a deadly radius of 10 meters (33 feet) upon impact.^[27]
- A high-explosive dual-purpose grenade capable of penetrating 50 mm (2.0 in) of steel armor.^[27]
- An inert cartridge for loading exercise.^[27]

40×56mm (Japan)

The standard adopted by the Japan Ground Self-Defense Force in its Type 96 grenade launcher. It is manufactured by Daikin Industries and the '40 mm anti-personnel light armour-piercing round' (ammunition length 112 mm, weight 371 g) is a multi-purpose grenade with a moulded explosive charge. There is also a '40 mm training round' available. ^[53] ^[54]^[55]

Caseless ammunition

40 mm VOG-25 (Russia)

40 mm VOG-25 (Russian Cyrillic: *BOF-25*) (GRAU-Index: 7P17 (Russian Cyrillic: *7П17*)) is a unique type of 40 mm grenade designed in the Soviet Union for hand-held grenade launchers, such as the Soviet GP-25 Kostyor and GP-30 Obuvka.^[56] Instead of a casing, the VOG-25 is caseless ammunition, featuring its propellant in an expansion chamber at the base of the projectile, functioning more like a mortar round than conventional cased ammunition.^[56]



VOG-25

Today it is used primarily by the Russian Armed Forces in weapons such as the GP-34, BG-15 Mukha and RG-6. Several types exist but the most common version is the default VOG-25 high-explosive version.^[56]

The VOG-25 is 103 mm (4.1 in) long, weighs 250 g (8.8 oz), and features a 48 g (1.7 oz) explosive charge. It has a muzzle velocity of 76 m/s (250 ft/s) and will self-destruct after 14 seconds.^[56]

40 mm Metal Storm (Australia)

During its time (1994–2012), Metal Storm Limited in Australia designed several automatic caseless 40 mm grenade launcher systems^[57]^[58] based on their own caseless ammunition weapon design.^[59] Unlike common caseless ammunition and their weapon systems the Metal Storm design lacked a feeding magazine and instead stacked the projectiles in front of each other in the barrel with the propellant in between the projectiles.^[59] The system lacked moving parts and the propellant was electronically fired. The 40 mm grenades used in the systems were off the shelf existing warheads converted to function in the design.^[57]

See also

- 35 mm grenade
- 40×46mmSR Hellhound
- United States 40 mm grenades

Notes

- NEM = Net Explosive Mass
- Comp A5 = Composition A5. Comp A5 is Hexogen + graphite according to Swedish manuals.
- NEM = Net Explosive Mass
- NEM = Net Explosive Mass

- e. Comp A3 = Composition A3. Comp A3 is Hexogen + wax according to Swedish manuals.
- f. Comp A5 = Composition A5. Comp A5 is Hexogen + graphite according to Swedish manuals.
- g. NEM = Net Explosive Mass
- h. NEM = Net Explosive Mass
- i. NEM = Net Explosive Mass

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